

N00 - Companion Notebook Index

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1 N00 — Companion Notebook Index

1.1 Engineering Quantum Interactions: From Effective Models to Validated Physical Predictions

This index is the entry point to the article’s computational companion. Each notebook is a self-contained computational lesson that links a microscopic model to an effective description, states the approximation domain, validates physical predictions with QuTiP, and exports the numerical data and figures used in the article or Supplemental Material.

The compact identifiers are permanent resource labels. Files with the same stem (`.ipynb`, `.html`, and `.pdf`) are alternate views of the same resource. `Nxx_MFy` identifies an article-scale figure; `Nxx_Fy` identifies a notebook-only figure.

1.2 Resource map

ID	Notebook	Scientific scope	Article-scale figures
N01	Beyond-RWA Mediated Interactions	Counter-rotating virtual paths, complete second-order interactions, Schrieffer–Wolff encoding and reconstruction, and QuTiP validation.	N01_MF1; N01_MF2
N02	Spin-Chain Quantum Bus	Mode- and resolvent-based virtual transport, retained-mode design, state transfer, leakage, and channel diagnostics.	N02_MF1; N02_MF2
N03	Floquet-Controlled Tunnelling	Bessel-renormalized hopping, coherent destruction of tunnelling, quasienergy gauge, micromotion, and Sambe-space checks.	N03_MF1; N03_MF2

ID	Notebook	Scientific scope	Article-scale figures
N04	Lossy Bosonic Mediator	Open-system adiabatic elimination, induced coherent and dissipative interactions, reconstruction, physicality, and memory.	N04_MF1; N04_MF2
N05	Three-Photon Rabi Resonance	Shifted three-photon resonance, cubic coupling, virtual-state reconstruction, and predictive-time validation.	N05_MF1
N06	Three-Level Quantum Mixers	Quantized Raman conversion, signed detunings, three-level topology, Stark-shifted resonance, and pair processes.	N06_MF1; N06_MF2
N07	Langevin, Liouville-Space, and Dynamical Maps	Heisenberg–Langevin equations, Gaussian channels, input–output fields, Liouville-space maps, physicality, and memory.	N07_F11
N08	Quantum Rabi Regime Map	Rotating-wave, dispersive, displaced-oscillator, multiphoton, and critical effective descriptions of the quantum Rabi model.	N08_MF1; N08_MF2; N08_MF3

1.3 How to use the collection

1. Create the shared environment from the repository root with `conda env create -f environment.yml`.
2. Activate it with `conda activate engineering-quantum-interactions`.
3. Open JupyterLab, enter one notebook directory, and run all cells from a fresh kernel.
4. Compare the generated `figures/manuscript/` and `data/` files with the manifest.

All scientific simulations use QuTiP as the quantum-dynamics implementation. NumPy, SciPy, pandas, and Matplotlib support numerical analysis, tabulation, and plotting.

```
[1]: from pathlib import Path
import json, platform, sys
import matplotlib, numpy, pandas, qutip, scipy

HERE = Path.cwd().resolve()
if HERE.name == 'N00_Companion-Notebook-Index':
    repo = HERE.parents[1]
elif (HERE / 'notebooks').is_dir():
    repo = HERE
else:
    raise RuntimeError('Run N00 from its directory or from the repository root.
↳')

manifest = pandas.read_csv(repo / 'notebooks' / 'NOTEBOOK_MANIFEST.csv')
for row in manifest.itertuples(index=False):
    source = repo / row.relative_directory / row.ipynb
    if not source.is_file():
        raise FileNotFoundError(source)

environment = {
    'python': sys.version.split()[0],
    'platform': platform.platform(),
    'qutip': qutip.__version__,
    'numpy': numpy.__version__,
    'scipy': scipy.__version__,
    'pandas': pandas.__version__,
    'matplotlib': matplotlib.__version__,
}
display(manifest[['resource_id', 'title', 'ipynb', 'pdf']])
print(json.dumps(environment, indent=2))
print(f'Validated {len(manifest)} companion notebook sources.')
```

	resource_id	title \
0	N00	Companion Notebook Index
1	N01	Beyond-RWA Mediated Interactions
2	N02	Spin-Chain Quantum Bus
3	N03	Floquet-Controlled Tunnelling
4	N04	Lossy Bosonic Mediator
5	N05	Three-Photon Rabi Resonance
6	N06	Three-Level Quantum Mixers
7	N07	Langevin, Liouville-Space, and Dynamical Maps
8	N08	Quantum Rabi Regime Map

	ipynb \
0	N00_Companion-Notebook-Index.ipynb
1	N01_Beyond-RWA-Mediated-Interactions.ipynb
2	N02_Spin-Chain-Quantum-Bus.ipynb
3	N03_Floquet-Controlled-Tunnelling.ipynb

```

4         N04_Lossy-Bosonic-Mediator.ipynb
5     N05_Three-Photon-Rabi-Resonance.ipynb
6         N06_Three-Level-Quantum-Mixers.ipynb
7         N07_Langevin-Liouville-Maps.ipynb
8         N08_Quantum-Rabi-Regime-Map.ipynb

```

```

                                pdf
0         N00_Companion-Notebook-Index.pdf
1     N01_Beyond-RWA-Mediated-Interactions.pdf
2         N02_Spin-Chain-Quantum-Bus.pdf
3     N03_Floquet-Controlled-Tunnelling.pdf
4         N04_Lossy-Bosonic-Mediator.pdf
5     N05_Three-Photon-Rabi-Resonance.pdf
6     N06_Three-Level-Quantum-Mixers.pdf
7         N07_Langevin-Liouville-Maps.pdf
8         N08_Quantum-Rabi-Regime-Map.pdf

```

```

{
    "python": "3.11.9",
    "platform": "Windows-10-10.0.26200-SP0",
    "qutip": "5.2.0",
    "numpy": "2.3.2",
    "scipy": "1.16.1",
    "pandas": "3.0.6",
    "matplotlib": "3.10.5"
}

```

Validated 9 companion notebook sources.

1.4 Reproducibility and citation

The repository-level CITATION.cff, codemeta.json, .zenodo.json, FIGURE_DATA_MAP.csv, and reports/ directory provide citation metadata, software metadata, figure provenance, and audit evidence. Cite the article for the scientific framework and cite the archived software release when reusing code or data.